Администрирование локальных сетей

Лабораторная работа №14

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Цели и задачи работы

Цель лабораторной работы

Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

Выполнение лабораторной работы

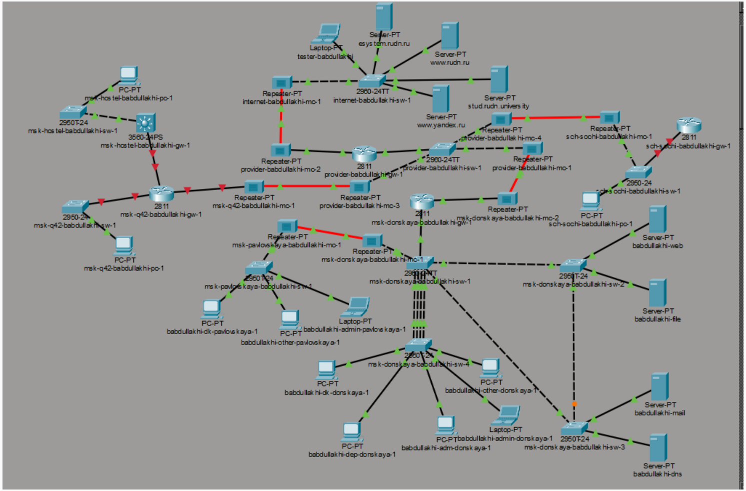


Рис. 1: Общая топология сети

Настройка VLAN на коммутаторе провайдера

```
Password:
provider-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-babdullakhi-sw-1(config)#int f0/3
provider-babdullakhi-sw-1(config-if)#sw mode trunk

provider-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

provider-babdullakhi-sw-1(config-if)#int f0/4
provider-babdullakhi-sw-1(config-if)#sw mode trunk
provider-babdullakhi-sw-1(config-if)#exit
provider-babdullakhi-sw-1(config)#vlan 5
provider-babdullakhi-sw-1(config-vlan)#name q42
provider-babdullakhi-sw-1(config-vlan)#int vlan 5
provider-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

provider-babdullakhi-sw-1(config-if)#no sh
provider-babdullakhi-sw-1(config-if)#vlan 6
provider-babdullakhi-sw-1(config-vlan)#name sochi
provider-babdullakhi-sw-1(config-vlan)#int vlan 6
provider-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-babdullakhi-sw-1(config-if)#no sh
provider-babdullakhi-sw-1(config-if)#exit
provider-babdullakhi-sw-1(config)#exit
provider-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
provider-babdullakhi-sw-1#
```

Настройка подинтерфейсов центрального маршрутизатора

```
msk-donskaya-babdullakhi-gw-1>en
Password:
msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#int f0/1.5
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#enc dot1q 5
msk-donskaya-babdullakhi-gw-1(config-subif)#ip addr 10.128.255.1 255.255.255.252
msk-donskaya-babdullakhi-gw-1(config-subif)#descr q42
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/1.6
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#enc dot1q 6
msk-donskaya-babdullakhi-gw-1(config-subif)#ip addr 10.128.255.5 255.255.255.252
msk-donskaya-babdullakhi-gw-1(config-subif)#descr sochi
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#
```

Рис. 3: Настройка подинтерфейсов на маршрутизаторе центральной площадки

Настройка статических маршрутов на центральном маршрутизаторе

```
msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-babdullakhi-gw-1(config)#ip route 10.130.0.0 255.255.0.0 10.128.255.6
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#int f0/1.5
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/1.6
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark q42
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#exit
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#wr mem
Building configuration...
```

Рис. 4: Настройка статических маршрутов и ACL на маршрутизаторе центральной площадки

Настройка маршрутизатора 42-го квартала

```
msk-q42-babdullakhi-gw-1(config-if)#int f0/0.201
msk-q42-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up

msk-q42-babdullakhi-gw-1(config-subif)#enc dot1q 201
msk-q42-babdullakhi-gw-1(config-subif)#ip addr 10.129.0.1 255.255.255.0
msk-q42-babdullakhi-gw-1(config-subif)#descr q42-main
msk-q42-babdullakhi-gw-1(config-subif)#int f1/0
msk-q42-babdullakhi-gw-1(config-if)#no shu

msk-q42-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

msk-q42-babdullakhi-gw-1(config-if)#int f1/0.202
msk-q42-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

msk-q42-babdullakhi-gw-1(config-subif)#enc dot1q 202
msk-q42-babdullakhi-gw-1(config-subif)#ip addr 10.129.1.1 255.255.255.0
msk-q42-babdullakhi-gw-1(config-subif)#descr q42-management
msk-q42-babdullakhi-gw-1(config-subif)#exit
msk-q42-babdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-babdullakhi-gw-1(config)#
msk-q42-babdullakhi-gw-1(config)#exit
msk-q42-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
msk-q42-babdullakhi-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-babdullakhi-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-babdullakhi-gw-1(config)#exit
msk-q42-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Настройка коммутатора 42-го квартала

```
msk-q42-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-babdullakhi-sw-1(config)#int f0/24
msk-q42-babdullakhi-sw-1(config-if)#sw mode trunk

msk-q42-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

msk-q42-babdullakhi-sw-1(config-if)#exit
msk-q42-babdullakhi-sw-1(config)#int f0/1
msk-q42-babdullakhi-sw-1(config-if)#sw mode acc
msk-q42-babdullakhi-sw-1(config-if)#sw acc vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-babdullakhi-sw-1(config-if)#vlan 201
msk-q42-babdullakhi-sw-1(config-vlan)#name q42-main
msk-q42-babdullakhi-sw-1(config-vlan)#int vlan 201
msk-q42-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up

msk-q42-babdullakhi-sw-1(config-if)#no sh
msk-q42-babdullakhi-sw-1(config-if)#exit
msk-q42-babdullakhi-sw-1(config)#exit
msk-q42-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
```

Рис. 6: Настройка коммутатора сети 42-го квартала

Настройка ПК в сети 42-го квартала

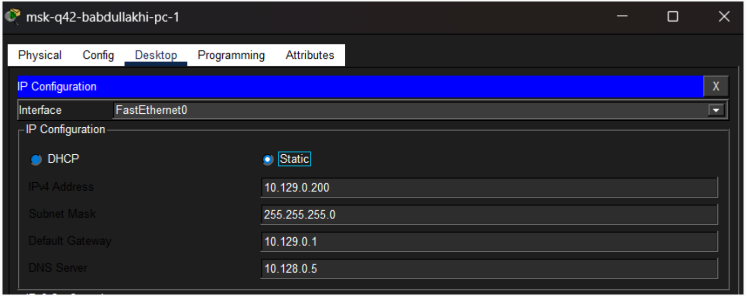


Рис. 7: Настройка IP-адреса на ПК сети 42-го квартала

Настройка шлюза сети общежития

```
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-babdullakhi-gw-1(config)#int g0/1
msk-hostel-babdullakhi-gw-1(config-if)#sw tr enc dot1q
msk-hostel-babdullakhi-gw-1(config-if)#sw mode trunk

msk-hostel-babdullakhi-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-hostel-babdullakhi-gw-1(config-if)#int f0/1
msk-hostel-babdullakhi-gw-1(config-if)#sw tr enc dot1q
msk-hostel-babdullakhi-gw-1(config-if)#sw mode trunk

msk-hostel-babdullakhi-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-hostel-babdullakhi-gw-1(config-if)#vlan 202
msk-hostel-babdullakhi-gw-1(config-vlan)#name q42-management
msk-hostel-babdullakhi-gw-1(config-vlan)#int vlan 202
msk-hostel-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

msk-hostel-babdullakhi-gw-1(config-if)#no shut
msk-hostel-babdullakhi-gw-1(config-if)#ip add 10.129.1.2 255.255.255.0
msk-hostel-babdullakhi-gw-1(config-if)#int vlan 301
msk-hostel-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-babdullakhi-gw-1(config-if)#no shut
msk-hostel-babdullakhi-gw-1(config-if)#ip add 10.129.129.1 255.255.255.0
msk-hostel-babdullakhi-gw-1(config-if)#vlan 301
msk-hostel-babdullakhi-gw-1(config-vlan)#name hostel-main
msk-hostel-babdullakhi-gw-1(config-vlan)#exit
msk-hostel-babdullakhi-gw-1(config)#ip routing
msk-hostel-babdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
msk-hostel-babdullakhi-gw-1(config)#exit
msk-hostel-babdullakhi-gw-1#
```

Настройка коммутатора сети общежития

```
msk-hostel-babdullakhi-sw-1>en
Password:
msk-hostel-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-babdullakhi-sw-1(config)#
msk-hostel-babdullakhi-sw-1(config)#int g0/1
msk-hostel-babdullakhi-sw-1(config-if)#sw mod tr
msk-hostel-babdullakhi-sw-1(config-if)#int f0/1
msk-hostel-babdullakhi-sw-1(config-if)#sw mode acc
msk-hostel-babdullakhi-sw-1(config-if)#sw acc vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-babdullakhi-sw-1(config-if)#vlan 301
msk-hostel-babdullakhi-sw-1(config-vlan)#name hostel-main
msk-hostel-babdullakhi-sw-1(config-vlan)#int vlan 301
msk-hostel-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-babdullakhi-sw-1(config-if)#no shut
msk-hostel-babdullakhi-sw-1(config-if)#exit
msk-hostel-babdullakhi-sw-1(config)#exit
msk-hostel-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-babdullakhi-sw-1#wr mem
Building configuration...
```

Рис. 9: Настройка коммутатора сети общежития

Настройка ПК сети общежития

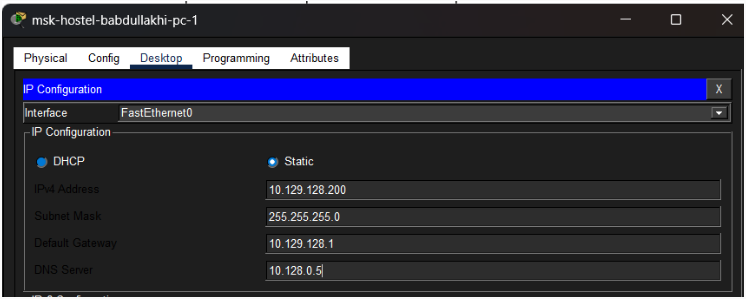


Рис. 10: Настройка IP-адреса на ПК сети общежития

Настройка маршрутизатора филиала в Сочи

```
Password:
sch-sochi-babdullakhi-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-babdullakhi-gw-1(config)#int f0/0
sch-sochi-babdullakhi-gw-1(config-if)#no shut

sch-sochi-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

sch-sochi-babdullakhi-gw-1(config-if)#int f0/0.6
sch-sochi-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up

sch-sochi-babdullakhi-gw-1(config-subif)#enc dot1q 6
sch-sochi-babdullakhi-gw-1(config-subif)#ip addr 10.128.255.6 255.255.255.252
sch-sochi-babdullakhi-gw-1(config-subif)#desc donskaya
sch-sochi-babdullakhi-gw-1(config-subif)#exit
sch-sochi-babdullakhi-gw-1(config)#int f0/0.401
sch-sochi-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

sch-sochi-babdullakhi-gw-1(config-subif)#enc dot1q 401
sch-sochi-babdullakhi-gw-1(config-subif)#ip addr 10.130.0.1 255.255.255.0
sch-sochi-babdullakhi-gw-1(config-subif)#int f0/0.402
sch-sochi-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up

sch-sochi-babdullakhi-gw-1(config-subif)#enc dot1q 402
sch-sochi-babdullakhi-gw-1(config-subif)#ip addr 10.130.1.1 255.255.255.0
sch-sochi-babdullakhi-gw-1(config-subif)#descr sochi-management
sch-sochi-babdullakhi-gw-1(config-subif)#int f0/0.401
sch-sochi-babdullakhi-gw-1(config-subif)#descr sochi-main
sch-sochi-babdullakhi-gw-1(config-subif)#exit
sch-sochi-babdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-babdullakhi-gw-1(config)#exit
sch-sochi-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Настройка коммутатора филиала в Сочи

```
Password:
sch-sochi-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-babdullakhi-sw-1(config)#int f0/23
sch-sochi-babdullakhi-sw-1(config-if)#sw mode trunk
sch-sochi-babdullakhi-sw-1(config-if)#int f0/24
sch-sochi-babdullakhi-sw-1(config-if)#sw mode trunk
sch-sochi-babdullakhi-sw-1(config-if)#vlan 6
sch-sochi-babdullakhi-sw-1(config-vlan)#name sochi
sch-sochi-babdullakhi-sw-1(config-vlan)#int vlan 6
sch-sochi-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

sch-sochi-babdullakhi-sw-1(config-if)#no shut
sch-sochi-babdullakhi-sw-1(config-if)#exit
sch-sochi-babdullakhi-sw-1(config)#
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

sch-sochi-babdullakhi-sw-1(config)#int f0/1
sch-sochi-babdullakhi-sw-1(config-if)#sw mode acc
sch-sochi-babdullakhi-sw-1(config-if)#sw acc vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-babdullakhi-sw-1(config-if)#vlan 401
sch-sochi-babdullakhi-sw-1(config-vlan)#name sochi-main
sch-sochi-babdullakhi-sw-1(config-vlan)#int vlan 401
sch-sochi-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

sch-sochi-babdullakhi-sw-1(config-if)#no sh
sch-sochi-babdullakhi-sw-1(config-if)#exit
sch-sochi-babdullakhi-sw-1(config)#exit
sch-sochi-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-babdullakhi-sw-1#wr mem
```

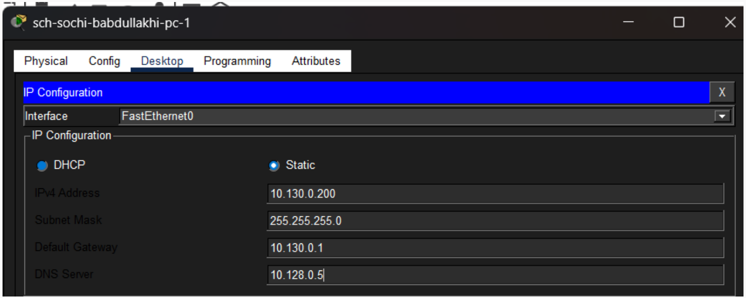
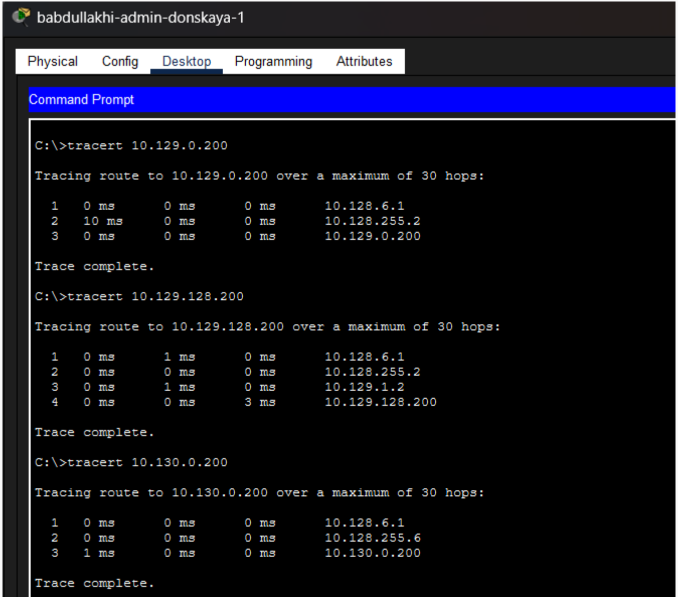


Рис. 13: Настройка IP-адреса на ПК филиала в Сочи

Проверка работы сети




```
C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.128.6.1
  2   10 ms     0 ms      0 ms      10.128.255.2
  3   0 ms      0 ms      0 ms      10.129.0.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  1   0 ms      1 ms      0 ms      10.128.6.1
  2   0 ms      0 ms      0 ms      10.128.255.2
  3   0 ms      1 ms      0 ms      10.129.1.2
  4   0 ms      0 ms      3 ms      10.129.128.200

Trace complete.

C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.128.6.1
  2   0 ms      0 ms      0 ms      10.128.255.6
  3   1 ms      0 ms      0 ms      10.130.0.200

Trace complete.
```

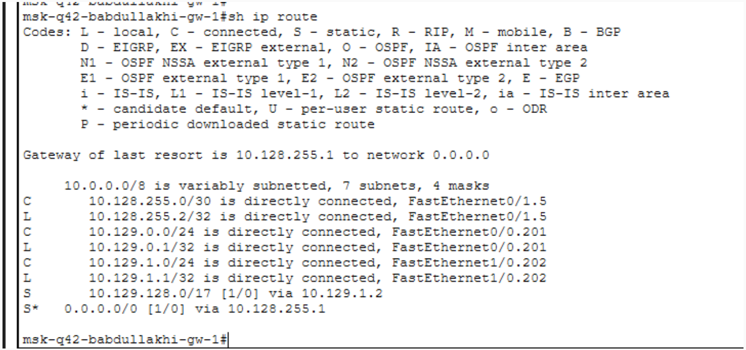
Таблица маршрутизации центрального маршрутизатора

```
msk-donskaya-babdullakhi-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
S       10.129.0.0/16 [1/0] via 10.128.255.2
S       10.130.0.0/16 [1/0] via 10.128.255.6
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1

msk-donskaya-babdullakhi-gw-1#
```



```
msk-q42-babdullakhi-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
C       10.129.1.0/24 is directly connected, FastEthernet1/0.202
L       10.129.1.1/32 is directly connected, FastEthernet1/0.202
S       10.129.128.0/17 [1/0] via 10.129.1.2
S*   0.0.0.0/0 [1/0] via 10.128.255.1

msk-q42-babdullakhi-gw-1#
```

Рис. 16: Таблица маршрутизации маршрутизатора 42-го квартала

Таблица маршрутизации шлюза общежития

```
msk-hostel-babdullakhi-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 2 subnets
C       10.129.1.0 is directly connected, Vlan202
C       10.129.128.0 is directly connected, Vlan301
S*   0.0.0.0/0 [1/0] via 10.129.1.1

msk-hostel-babdullakhi-gw-1#
```

Рис. 17: Таблица маршрутизации шлюза сети общежития

Таблица маршрутизации маршрутизатора Сочи

```
sch-sochi-babdullakhi-gw-1>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*   0.0.0.0/0 [1/0] via 10.128.255.5
```

Рис. 18: Таблица маршрутизации маршрутизатора филиала в Сочи

Выводы по проделанной работе

Вывод

В ходе лабораторной работы была настроена передача данных между центральной площадкой, сетью основного здания в 42-м квартале, сетью общежития и филиалом в Сочи.

Для организации взаимодействия были настроены VLAN, trunk-соединения, подинтерфейсы с инкапсуляцией dot1Q и статические маршруты.

Проверка с помощью команды tracert показала успешное прохождение маршрутов до удалённых узлов. Таблицы маршрутизации подтвердили наличие необходимых статических маршрутов и маршрутов по умолчанию.